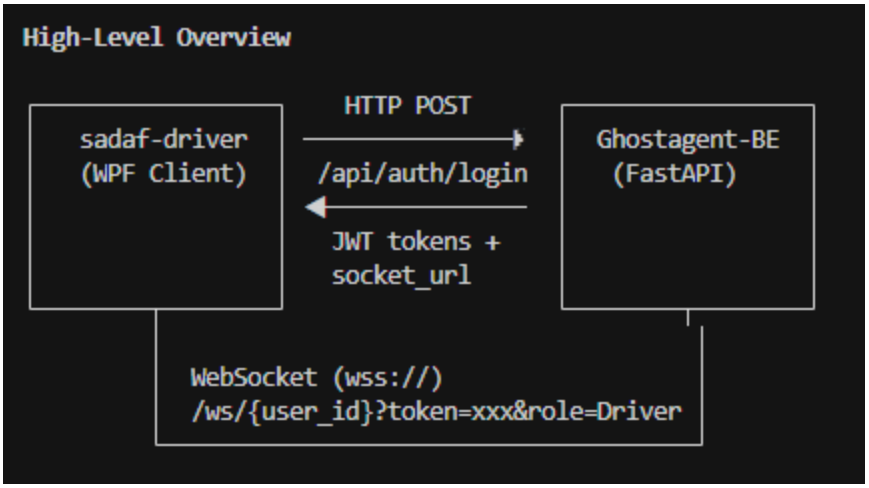
Driver

1. Overview

This document outlines the JSON payload structures that the Ghost Agent driver must send to the backend application over a WebSocket connection. The driver is responsible for initiating this connection and streaming events throughout the lifecycle of a customer call.

2. Connection



---

3. General Message Format

All messages sent from the driver to the frontend **must** be a JSON object containing an `event` field. This field determines the structure of the rest of the payload.

```
{
  "event": "event_name",
  "user_id": "<user_id>",
  ...payload
}
```

All timestamps **must** be in **ISO 8601 format** (e.g., `2025-10-06T11:45:10.123Z`).

4. Event Payloads

The following are the defined events and their corresponding payloads.

4.1 `incoming_call_ringing`

Sent by the driver when the it detects an incoming call that is being routed to the agent

Field	Type	Description
event	string	"incoming_call_ringing"
call_id	string	A unique identifier for the incoming call, generated by the IVR or driver.
customer	object	Basic information about the incoming caller.
received_at	string	The timestamp when the call started ringing.

```
{
  "event": "incoming_call_ringing",
  "call_id": "call_xyz789",
  "customer": {
    "phone_e164": "+919876543210"
  },
  "received_at": "2025-10-06T15:30:00.123Z"
}
```

4.2 `call_started`

Sent **once** at the very beginning of a call when the connection is established.

Field	Type	Description
event	string	"call_started"
agent_id	string	The unique ID of the ElevenLabs agent handling the call.
conversation_id	string	The unique ID for the conversation session from ElevenLabs.
session_id	string	A unique UUID v4 generated by the driver for this entire interaction. Used to identify the WebSocket session.
employee	object	The profile of the logged-in sales representative, fetched by the driver from the /auth/me endpoint.
customer	object	Information about the customer.
started_at	string	The timestamp when the call officially started.

```
{
  "event": "call_started",
  "agent_id": "agt_X5gH9kLpW2sR7tYq",
  "conversation_id": "conv_aB8fG3jLpW2sR7tY",
  "session_id": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "employee": {
    "id": "c7a4a5d8-2b8e-4f7d-9e6a-1b2c3d4e5f6g",
    "email": "test123@test.com",
    "first_name": "test",
    "last_name": "tester"
  },
  "customer": {
    "name": "John Doe",
    "customerId": "CUST-1024",
    "phone_e164": "+919876543210",
    "email": "john.doe@example.com"
  },
```

```
  "started_at": "2025-10-06T11:45:10.123Z"
}
```

4.3 `transcript_turn`

Sent whenever a new "turn" (a complete utterance by one speaker) is finalized by the transcription service.

Field	Type	Description
event	string	"transcript_turn"
conversation_id	string	The unique ID for the conversation session from ElevenLabs.
turn_id	string	A unique identifier for this specific turn of the conversation.
speaker	string	The speaker of the text. Enum: "caller" or "CSR".
text	string	The transcribed text of the utterance.
timestamp	string	timestamp of this transcript

```
{
  "event": "transcript_turn",
  "conversation_id": "conv_aB8fG3jLpW2sR7tY",
  "turn_id": "turn_001",
  "speaker": "caller",
  "text": "Hi, my name is Sophia. My email ID is sophiawhite@
email.com. My order is not here."
  "timestamp": "2025-10-06T11:45:18.456Z"
}
```

4.4 `mcp_tool_call`

Sent as soon as the ElevenLabs agent calls a tools and sends a tool call result

Field	Type	Description
event	string	"mcp_tool_call”

conversation_id	string	The unique ID for the conversation session from ElevenLabs.
tool_call_id	string	A unique identifier for this specific tool call.
tool_name	string	The name of the tool being called (e.g., get_order_overview).
parameters	object	The parameters being passed to the tool.
timestamp	string	The timestamp when the tool call was initiated.
state	string	shows the state of tool call, success state
result		mcp tool call result


```
{
  "event": "mcp_tool_call",
  "conversation_id": "conv_aB8fG3jLpW2sR7tY",
  "tool_call_id": "tool_call_abc",
  "tool_name": "get_customer_info",
  "parameters": {
      "identifier": "sophiawhite@email.com"
    },
  "timestamp": "2025-10-30T05:44:14.564702+00:00",
  "state": "success",
  "result": [
      {
        "type": "text",
        "text": "👤 **Customer Information**\nName: Sophia Wh
ite\nEmail: sophiawhite@email.com\nPhone: 555-0114\nAddress:
357 Cherry Ave, Columbus, OH 43085\n",
        "annotations": null,
        "meta": null
      }
    ]
}
```


4.8 `agent_thinking`

The agent thinking is designed to give the human agent a clear, concise summary of what the AI understood and what it's about to do. The content of the `thinking` is generated by the agent.

Field	Type	Description
event	string	"agent_thinking"
conversation_id	string	The unique ID for the conversation session from ElevenLabs.
thinking	string	A structured object representing the AI's intention and next action.
timestamp	string	The timestamp when the driver generated this event.

```
{
  "event": "agent_thinking",
  "conversation_id": "conv_aB8fG3jLpW2sR7tY",
  "thinking": "Customer provided their name and email addres
s, and mentioned an issue with their order. I will retrieve c
ustomer information using the email address.",
  "timestamp": "2025-10-06T12:35:09.500Z"
}
```

Sent once when the conversation is terminated for any reason.

Field	Type	Description
event	string	"call_ended"
conversation_id	string	The unique ID for the conversation session from ElevenLabs.
session_id	string	The unique ID for the WebSocket session.
reason	string	The reason the call ended. Enum: "agent_hung_up", "customer_hung_up", "completed", "error".
ended_at	string	The timestamp when the call ended.
duration_secs	integer	The total duration of the call in seconds.

```
{
  "event": "call_ended",
  "conversation_id": "conv_aB8fG3jLpW2sR7tY",
  "session_id": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "reason": "customer_hung_up",
  "ended_at": "2025-10-06T11:58:30.999Z",
  "duration_secs": 800
}
```

4.7 `error_occured`

A generic event for handling unexpected, non-fatal errors from the driver or its connected services (AssemblyAI, ElevenLabs).This provides a way to display a clear error message to the agent in the UI (e.g., "Transcription service disconnected") without crashing the entire session.

Field	Type	Description
event	string	"error_occurred"
session_id	string	The unique ID for the WebSocket session.
source	string	The source of the error. Enum: "driver", "transcription", "elevenlabs".
message	string	A user-friendly error message.
details	string	Optional technical details for debugging.
timestamp	string	The timestamp when the driver generated this event.

```
{
  "event": "error_occurred",
  "session_id": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "source": "transcription",
  "message": "Connection to transcription service was lost.Tr
anscript may be incomplete.",
  "details": "WebSocket connection timed out.",
```

```
  "timestamp": "2025-10-06T12:40:05.500Z"
}
```